CHE210: ***DISCUSSION SESSION 12*** **NAME:**___________________________________

APRIL 20, 2020

The ethanol dehydrogenation reaction found below is carried out with the feed entering at 300°C. The feed contains 90.0 mole% ethanol and the balance acetaldehyde and enters the reactor at a rate of 150 mol/s. To keep the temperature from dropping too much and thereby decreasing the reaction rate to an unacceptably low level, heat is transferred to the reactor. When the heat addition rate is 2440 kW, the outlet temperature is 253°C. Calculate the fractional conversion of ethanol achieved in the reactor.

$$C_2H_5OH \rightarrow CH_3CHO + H_2$$

Useful Data:

$$\Delta H^{\circ}_{f,E} = -235.31 \frac{kJ}{mol}$$

$$\Delta H^{\circ}_{f,A} = -166.2 \frac{kJ}{mol}$$

$$C_{p,E}(T°C) = 0.06134 + 1.572 \times 10^{-4}\, T - 8.749 \times 10^{-8}\, T^2 + 1.983 \times 10^{-11}\, T^3$$

$$C_{p,A}(T°C) = 0.05048 + 1.326 \times 10^{-4}\, T - 8.05 \times 10^{-8}\, T^2 + 2.38 \times 10^{-11}\, T^3$$

$$C_{p,H}(T°C) = 0.02884 + 7.65 \times 10^{-8}\, T + 3.288 \times 10^{-9}\, T^2 - 8.698 \times 10^{-13}\, T^3$$